

Chapter 4

Linear Algebra

4.1 Free Modules

4.1.1 Linear Independence

One of the key definitions in linear algebra is that of a basis.

Definition 4.1.1. Let M be a left R -module, and let $S \subseteq M$.

- (1) S is said to be *linearly dependent* if there are $m_1, \dots, m_n \in S$ and $\alpha_1, \dots, \alpha_n \in R$ not all 0 such that

$$\alpha_1 m_1 + \dots + \alpha_n m_n = 0.$$

If S is not linearly dependent, we say it is *linearly independent*.

- (2) S is said to be *spanning* if $\langle S \rangle = M$.
- (3) S is said to be a *basis* if S is linearly independent and spanning.

We say elements $m_1, \dots, m_n$ are linearly independent if $\{m_1, \dots, m_n\}$ is linearly independent. If $S \subseteq T$ and T is linearly independent, then S is linearly independent as well. Note that linear independence is a *finite* property: a set S is linearly independent if and only if every finite subset of S is linearly independent.

Proposition 4.1.2. Let M be a left R -module and $S \subseteq M$. Then S is linearly independent if and only if

$$\sum_{s \in S} \alpha_s s = \sum_{s \in S} \beta_s s \iff \alpha_s = \beta_s \forall s.$$

In particular, S is a basis if and only if every $m \in M$ has a unique representation in the form $\sum_{s \in S} \alpha_s s$.

Proof. Suppose that S is linearly independent and $\sum_{s \in S} \alpha_s s = \sum_{s \in S} \beta_s s$. Then $\sum_{s \in S} (\alpha_s - \beta_s) s = 0$, so linear independence gives $\alpha_s - \beta_s = 0$ for all $s \in S$. Conversely, if representations as linear combinations of S are unique and $\sum_{s \in S} \alpha_s s = 0$, then $\sum_{s \in S} \alpha_s s = \sum_{s \in S} 0s$, so that $\alpha_s = 0$ for all $s \in S$ and S is linearly independent. ■

Recall that we constructed the free module on S by taking the direct sum $R^{\oplus S}$. This free module F satisfies the universal property that any set map $S \rightarrow M$ lifts to a module map $F \rightarrow M$. By abstract nonsense, any two modules satisfying this universal property are isomorphic, but of course two distinct modules may satisfy the same universal property. The knowledge of bases allows us to determine precisely the R -modules that are free over some set B .

Proposition 4.1.3. Let M be a left R -module, $B \subseteq M$, and let $\phi: R^{\oplus B} \rightarrow M$ be the induced map.

- (1) B is linearly independent if and only if ϕ is injective.
- (2) B is spanning if and only if ϕ is surjective.

In particular, $M \cong F$ is a free module on B if and only if B is a basis.

Proof.

- (1) Suppose that B is linearly independent, and that $(\alpha_b)_{b \in B} \in R^{\oplus B}$ lies in $\ker \phi$. Then $\sum_{b \in B} \alpha_b b = 0$, and so $\alpha_b = 0$ for all $b \in B$. Thus, ϕ is injective. Conversely, if ϕ is injective and $\sum_{b \in B} \alpha_b b = 0$ with almost all $\alpha_b = 0$, then $(\alpha_b)_{b \in B} \in \ker \phi$, so $\alpha_b = 0$ for all b .
- (2) If B is spanning and $m \in M$, we can write $m = \sum_{b \in B} \alpha_b b$, so that $m = \phi((\alpha_b)_{b \in B})$. Conversely, if ϕ is surjective and $m \in M$, then $m = \phi((\alpha_b)_{b \in B})$, so that $m = \sum_{b \in B} \alpha_b b$ and B is spanning. ■

Remark 4.1.4. We may hope that we can recover a set B from the free module on B . Unfortunately, this is not true. For instance, let $M = R^{\oplus \mathbb{N}}$, and fix an isomorphism $(\rho_1, \rho_2) : M \cong M \oplus M$. Write $S = \text{End}_R(M)$, so that we have an isomorphism of abelian groups

$$S = \text{Hom}_R(M, M) \cong \text{Hom}_R(M, M \oplus M) \cong \text{Hom}_R(M, M) \oplus \text{Hom}_R(M, M) = S \oplus S$$

given by $\phi \mapsto (\rho_1 \phi, \rho_2 \phi)$. It is not hard to check that this is in fact an isomorphism of right S -modules as well. Thus, $\{1\} \in S$ and $\{(1, 0), (0, 1)\} \in S \oplus S$ are both maximal linearly independent sets.

For linear algebra, we want to deal with rings where we do not run into the problem above: $R^{\oplus S} \cong R^{\oplus T}$ should force $|S| = |T|$. This is the purpose of our next definition.

Definition 4.1.5. A ring R is said to have *left invariant basis number (LIBN)* if $R^n \cong R^m$ as left R -modules if and only if $n = m$, where $n, m \in \mathbb{N}$. Similarly, R is said to have *right invariant basis number (RIBN)* if $R^n \cong R^m$ as right R -modules if and only if $n = m$.

As it will turn out, a ring has left IBN if and only if it has right IBN, see . We'll generally say a ring has IBN if it has left IBN. Of course, IBN only guarantees that bases are well-defined when one is finite, and there are many conditions similar to IBN that allow us to make stronger claims, see .

Example. Finite rings have IBN. Let R be finite, and $n, m \in \mathbb{N}$. Then module homomorphisms $R^{\oplus n} \rightarrow R$ are in bijection with set maps $[n] \rightarrow R$, of which there are $|R|^n$. Thus, if $R^{\oplus n} \cong R^{\oplus m}$, then $|R|^n = |R|^m$, and so $n = m$, as $|R| < \infty$.

Example. Left Noetherian rings have IBN. If $R^{\oplus n} \cong R^{\oplus m}$ with $n < m$ and $\pi : R^{\oplus n} \cong R^{\oplus m} \rightarrow R^{\oplus n}$ is projection onto the first n coordinates, then π is a non-injective surjection. Now, $\ker \pi \subseteq \ker \pi^2 \subseteq \ker \pi^3 \subseteq \dots$ must stabilize, say $\ker \pi^k = \ker \pi^{k+1}$. But, since π is surjective, we can find $m \in M$ such that $\pi^k(m)$ is a nonzero element of $\ker \pi$ (recalling that $\ker \pi$ is nonzero), so that $\pi^{k+1}(m) = 0$ while $\pi^k(m) \neq 0$, a contradiction. Thus, no such π can exist.

We now turn to showing that all commutative rings have IBN. We start by showing it for fields.

Lemma 4.1.6. *Let M be a left R -module and $S \subseteq M$ be linearly independent. Then we can find a maximal linearly independent set containing S .*

Proof. We use Zorn's lemma. Let $(S_i)_{i \in I}$ be a chain of linearly independent subsets containing S , and set $T = \bigcup_{i \in I} S_i$. If we show that T is linearly independent, then Zorn's lemma yields the result. Suppose that $s_1, \dots, s_n \in T$, and take $s_j \in S_{j_i}$. Since we have a chain, one of these contains the rest, say S_{j_n} . Since S_{j_n} is linearly independent, so is $\{s_1, \dots, s_n\}$. Since these were arbitrary elements of T , T is linearly independent as well. ■

Theorem 4.1.7. *Let D be a division ring.*

- (1) *Every D -module is free. Moreover, every maximal linearly independent set is a basis.*
- (2) *If $S \subseteq M$ is linearly independent and $B \subseteq M$ spans M , then $|S| \leq |B|$.*
- (3) *If $D^{\oplus S} \cong D^{\oplus T}$, then $|S| = |T|$.*

In particular, D has IBN, and these results hold for fields.

Proof.

- (1) Let M be a D -module, and let $S \subseteq M$ be a maximal linearly independent set. Suppose that $m \in M$ is nonzero. If $m \in S$, then clearly $m \in \langle S \rangle$. Otherwise, $S \cup \{m\}$ is linearly independent, say $\alpha_m m + \sum_{s \in S} \alpha_s s = 0$ with not all coefficients equal to 0. Since S is linearly independent, we must have that $\alpha_m \neq 0$. Thus, $m = -\sum_{s \in S} \alpha_m^{-1} \alpha_s s \in \langle S \rangle$, so $M = \langle S \rangle$ is free.

- (2) We'll construct an injection $S \hookrightarrow B$. We proceed by transfinite induction: fix a well-ordering on S , take $s \in S$, and suppose by induction we have a map $f : \{t \in S : t < s\} \rightarrow B$ such that $X = \{t \in S : t < s\} \cup (B \setminus \text{im } f)$ is a spans M (taking f to be the empty injection establishes the base case). Then $s \in \langle X \rangle$, say

$$s = \sum_{t < s} \alpha_t t + \sum_{b \in B \setminus \text{im } f} \beta_b b.$$

Since B is linearly independent, some β_b must be nonzero, say β_{b_0} . Define

$$f'(t) = \begin{cases} f(t) & t < s \\ s_0 & t = s \end{cases}.$$

We claim that f' satisfies the conditions of the inductive hypothesis. Since

$$b_0 = \beta_{b_0}^{-1} \left(s - \sum_{t < s} \alpha_t t - \sum_{b \neq b_0 \in B \setminus \text{im } f} \beta_b b \right) \in \langle X' \rangle,$$

we have that $M = \langle X \rangle \leq \langle X' \rangle$. Thus, X' spans M .

- (3) $|S| \leq |T|$ and $|T| \leq |S|$

■

The key to the next proof is that every commutative ring has a quotient that is a field¹.

Theorem 4.1.8. *Let R be a commutative ring. Then $R^{\oplus X} \cong R^{\oplus Y}$ if and only if $|X| = |Y|$.*

Proof. Let $\mathfrak{m}$ be a maximal ideal of R . We have a surjective map $R^{\oplus X} \rightarrow (R/\mathfrak{m})^{\oplus X}$ given by reducing modulo $\mathfrak{m}$. The kernel of this map is $\mathfrak{m}R^{\oplus X}$. Thus,

$$(R/\mathfrak{m})^{\oplus X} \cong R^{\oplus X} / \mathfrak{m}R^{\oplus X} \cong R^{\oplus Y} / \mathfrak{m}R^{\oplus Y} \cong (R/\mathfrak{m})^{\oplus Y}.$$

Since $R/\mathfrak{m}$ is a field, $|X| = |Y|$.

■

We end by developing a bit of the theory of linear independence in any module.

Proposition 4.1.9. *Let R be an integral domain, M be an R -module, and $B \subseteq M$ be a maximal linearly independent set, and let $S \subseteq B$ be linearly independent. Then $|S| \leq |B|$.*

Proof. Set $k = \text{Frac } R$, and let V be the localization $(R \setminus \{0\})^{-1}M$, which is naturally a k -module. We'll show that $S/1 = \{s/1 : s \in S\}$ is a linearly independent subset of V , and that $B/1 = \{b/1 : b \in B\}$ is a maximal linearly independent subset of V , so that this follows from the result for fields. It is indeed linearly independent: if $\sum_{i=1}^n \frac{\alpha_i}{\beta_i} \frac{s_i}{1} = 0$, then we may clear denominators to get

$$\sum_{i=1}^n \left(\alpha_i \prod_{j \neq i} \beta_j \right) \frac{s_i}{1} = 0.$$

This means there is $\gamma \in R \setminus 0$ such that

$$\gamma \sum_{i=1}^n \left(\alpha_i \prod_{j \neq i} \beta_j \right) s_i = 0,$$

so that $\alpha_i = 0$ for all i , since the β_j and γ are all nonzero.

Now, for any $m/\gamma \in V$, we have that $\{m\} \cup B$ is linearly dependent, say $\alpha_m m + \sum_{b \in B} \alpha_b b = 0$ with $\alpha_m \neq 0$. Then,

$$(\alpha_m \gamma) \frac{m}{r} + \sum_{b \in B} \alpha_b \frac{b}{1} = 0.$$

Thus, $B/1$ is a maximal linearly independent set of V , and so $V = k^{\oplus B}$. Since $S \subseteq V$ is linearly independent, $|S| \leq |B|$.

■

¹We are actually taking a tensor product, see

In particular, any two maximal linearly independent subsets have the same cardinality.

Definition 4.1.10. Let R be a ring.

- (1) If R has IBN, the *rank* of a free R -module $M \cong R^{\oplus n}$ is $\text{rk}_R M = n$.
- (2) If R is an integral domain, the *rank* of a module M is the size of a maximal linearly independent subset of M . We write $\text{rk}_R M$ for this number.
- (3) If R is a division ring, the *dimension* of a module M is $|B|$, where $M \cong R^{\oplus B}$. We write $\dim_R M = |B|$.

These values all agree in the relevant cases.

4.1.2 Matrices

We will use the theory of matrices to understand linear maps $R^{\oplus n} \rightarrow R^{\oplus m}$. We first note that there is more than just the structure of abelian groups on $\text{Hom}_R(R^{\oplus n}, R^{\oplus m})$: we have a natural left R -module structure given by $(\alpha\phi)(m) = \alpha \cdot \phi(m)$. We will show that there is an isomorphism $\text{Hom}_R(R^{\oplus n}, R^{\oplus m}) \cong \mathbb{M}_{m \times n}(R)$ that agrees with the module structure.

Let $B = \{b_1, \dots, b_n\}$ and $C = \{c_1, \dots, c_m\}$ be bases of M and N , and let $\phi : M \rightarrow N$ be a homomorphism. We can write

$$\phi(b_i) = \sum_{j=1}^m \alpha_{j,i} c_j.$$

We define

$$[\phi]_{B,C} = (\alpha_{j,i})_{j,i}.$$

This process is injective, since a map is completely determined by how it acts on a basis. Further, this is a homomorphism of left R -modules: if $[\psi]_{B,C} = (\beta_{j,i})_{j,i}$, then

$$(\phi + \psi)(b_i) = \phi(b_i) + \psi(b_i) = \sum_{j=1}^m \alpha_{j,i} c_j + \sum_{j=1}^m \beta_{j,i} c_j = \sum_{j=1}^m (\alpha_{j,i} + \beta_{j,i}) c_j.$$

Thus,

$$[\phi + \psi]_{B,C} = (\alpha_{j,i} + \beta_{j,i})_{j,i} = (\alpha_{j,i})_{j,i} + (\beta_{j,i})_{j,i} = [\phi]_{B,C} + [\psi]_{B,C}.$$

Further, for $\gamma \in R$,

$$(\gamma\phi)(b_i) = \gamma \cdot \phi(b_i) = \gamma \sum_{j=1}^m \alpha_{j,i} c_j = \sum_{j=1}^m (\gamma\alpha_{j,i}) c_j,$$

so that

$$[\gamma\phi]_{B,C} = (\gamma\alpha_{j,i})_{j,i} = \gamma(\alpha_{j,i})_{j,i} = \gamma[\phi]_{B,C}.$$

Now, the map ϕ can be recovered from its matrix $[\phi]_{B,C}$ via the composition

$$M \xrightarrow{\sim} R^{\oplus B} \xrightarrow{[\phi]_{B,C}} R^{\oplus C} \xrightarrow{\sim} N$$

where the map $[\phi]_{B,C}$ is matrix multiplication. Thus, every matrix yields a corresponding homomorphism², and every homomorphism corresponds to a unique matrix, so we have an isomorphism $\text{Hom}_R(M, N) \cong \mathbb{M}_{m \times n}(R)$.

We may now wonder how composition interacts with taking matrices. The answer is matrix multiplication: if B is a basis of M , C is a basis of N , and D is a basis of P , then we have a commutative diagram

$$\begin{array}{ccccc}
 & & N & & \\
 & \nearrow \phi & \updownarrow \wr & \nwarrow \psi & \\
 & & R^{\oplus C} & & \\
 M & \xrightarrow{\sim} R^{\oplus B} & \xrightarrow{[\phi]_{B,C}} R^{\oplus C} & \xrightarrow{[\psi]_{C,D}} R^{\oplus D} & \xrightarrow{\sim} P \\
 & \searrow \psi\phi & \xrightarrow{[\psi\phi]_{B,D}} & & \\
 & & & &
 \end{array}$$

²We are sweeping under the rug the fact that multiplying by a matrix is indeed a homomorphism, a fact that is routine to check

Thus, multiplication by $[\psi\phi]_{B,D}$ and $[\psi]_{C,D}[\phi]_{B,C}$ agree as maps $R^{\oplus B} \rightarrow R^{\oplus D}$. Evaluating these on the elements of B shows that they are in fact the same matrix.

Now, free modules can have multiple distinct bases, and so we may want to find out how matrices with respect to these bases interact. The solution to this follows from the fact that composition corresponds to matrix multiplication: we have a commutative diagram

$$\begin{array}{ccccccc}
 & & \phi & & & & \\
 & \nearrow & & \searrow & & \nearrow & \\
 M & \xrightarrow{\quad\quad\quad} & M & \xrightarrow{\quad\phi\quad} & N & \xrightarrow{\quad\quad\quad} & N \\
 \downarrow \wr & & \downarrow \wr & & \downarrow \wr & & \downarrow \wr \\
 R^{\oplus B'} & \xrightarrow{[\text{id}_M]_{B',B}} & R^{\oplus B} & \xrightarrow{[\phi]_{B,C}} & R^{\oplus C} & \xrightarrow{[\text{id}_N]_{C,C'}} & R^{\oplus C'} \\
 & \searrow & & \nearrow & & \searrow & \\
 & & [\phi]_{B',C'} & & & &
 \end{array}$$

so that

$$[\phi]_{B',C'} = [\text{id}_N]_{C,C'}[\phi]_{B,C}[\text{id}_M]_{B',B}.$$

Further,

$$[\text{id}_M]_{B',B}[\text{id}_M]_{B,B'} = [\text{id}_M]_{B,B} = I_n,$$

so that $[\text{id}_M]_{B',B}$ (and similarly $[\text{id}_N]_{C,C'}$) is invertible. In the case we are considering endomorphisms $\phi : M \rightarrow M$, we write $[\phi]_B = [\phi]_{B,B}$.

Often, given a map $\phi : M \rightarrow M$, we will want to find a basis B of M such that the matrix $[\phi]_B$ is of a particularly nice form. This will be the point of study of much of this chapter. For now, we present some elementary results that will end up being useful later.

Definition 4.1.11. Let M be a left R -module, $N \leq M$ be a submodule, and $\phi : M \rightarrow M$ be a homomorphism. We say N is ϕ -invariant if $\phi(N) \subseteq N$.

If $N \leq M$ is ϕ -invariant, then we get maps $\phi|_N : N \rightarrow N$ by restriction and $\phi/N : M/N \rightarrow M/N$ induced by $M \xrightarrow{\phi} M \rightarrow M/N$. We'll be able to decompose the matrix of ϕ in terms of the matrices of these maps.

Proposition 4.1.12. Let M be a left R -module.

- (1) If $M = M_1 \oplus \cdots \oplus M_n$, B_i is a basis of M_i , and the M_i are ϕ -invariant, then $B = B_1 \cup \cdots \cup B_n$ is a basis of M , and

$$[\phi]_B = \begin{pmatrix} [\phi|_{M_1}]_{B_1} & & \\ & \ddots & \\ & & [\phi|_{M_n}]_{B_n} \end{pmatrix}.$$

- (2) If $N \leq M$ is ϕ -invariant, B is a basis of N , and $\{c + M : c \in C\}$ is a basis of M/N , then $B \cup C$ is a basis of M , and

$$[\phi]_{B \cup C} = \begin{pmatrix} [\phi|_N]_B & * \\ 0 & [\phi/N]_C \end{pmatrix}.$$

Proof.

- (1) The matrix representation is clear once we show that B is a basis. Since every element of M is a sum of elements of M_i , and every element of M_i is a sum of elements of B_i , B spans M . If

$$\sum_{i=1}^n \sum_{b \in B_i} \alpha_b b = 0,$$

then the fact that we have a direct sum means $\sum_{b \in B_i} \alpha_b b = 0$ for all i , and so linear independence gives $\alpha_b = 0$ for all b . Thus, B is indeed a basis.

- (2) We first show that $B \cup C$ is a basis. Suppose that $\sum_{b \in B} \alpha_b b + \sum_{c \in C} \alpha_c c = 0$. Reducing mod N , $\sum_{c \in C} \alpha_c (c + N) = 0 + N$, so that $\alpha_c = 0$ for all $c \in C$. Thus, $\sum_{b \in B} \alpha_b b = 0$, and so $\alpha_b = 0$ for all $b \in B$ by linear independence, meaning $B \cup C$ is linearly independent. If $m \in M$, we can write $m + N = \sum_{c \in C} \beta_c (c + N)$, so that $m - \sum_{c \in C} \beta_c c \in N$. Since B spans N , we can write $m - \sum_{c \in C} \beta_c c = \sum_{b \in B} \beta_b b$. Thus, $B \cup C$ spans M , so $B \cup C$ is a basis.

Since N is B -invariant, the left column of the matrix representation is clear. For any $c \in C$, reducing $\phi(c)$ modulo N shows the right column.

In particular, if we have a direct sum decomposition of vector spaces $V = V_1 \oplus \cdots \oplus V_n$, then $\dim V = \sum_{i=1}^n \dim V_i$. ■

4.2 Vector Spaces

4.2.1 Rank-Nullity

We will primarily be concerned with *finite-dimensional* vector spaces over k : vector spaces V such that $\dim_k V < \infty$. By Theorem 4.1.7, a vector space is finite-dimensional if and only if it has a finite basis if and only if it has a finite spanning set. Further, since every vector space is free, they are completely determined by their dimension, a fact that is not true for general rings. For instance, if $\dim V = 0$, then $V \cong k^{\oplus 0} \cong 0$.

Proposition 4.2.1. *Let V be a vector space over k and $W_1, W_2 \leq V$.*

- (1) *If V is finite-dimensional, then so is W_1 , and $\dim W_1 \leq \dim V$.*
- (2) *If W_1 and W_2 are finite-dimensional, then so are $W_1 + W_2$ and $W_1 \cap W_2$, and*

$$\dim W_1 + W_2 + \dim W_1 \cap W_2 = \dim W_1 + \dim W_2.$$

Proof.

- (1) Let B be a (finite) basis for V , and let C be a basis for W_1 . Then in particular C is linearly independent, so $|C| \leq |B|$ by Theorem 4.1.7. Thus, C is finite and $\dim W_1 \leq \dim V$.
- (2) Since $W_1 \cap W_2 \leq W_1$, it is finite-dimensional by (1). Let B be a basis for $W_1 \cap W_2$, and extend B to a basis B_1 of W_1 and B_2 of W_2 . Note that $B_1 \setminus B$ and $B_2 \setminus B$ are disjoint, as they lie in the disjoint sets $W_1 \setminus W_1 \cap W_2$ and $W_2 \setminus W_1 \cap W_2$. Thus, $|B_1 \cup B_2| = |B_1| + |B_2| - |B|$, so it suffices to show that $B_1 \cup B_2$ is a basis for $W_1 + W_2$.

Any element of $W_1 + W_2$ can be written in the form $w_1 + w_2$ for $w_i \in W_i$. Since w_i can be written as a linear combination of elements of B_i , this means that $B_1 \cup B_2$ spans $W_1 + W_2$. Now, suppose we have a relation $\sum_{b \in B_1 \cup B_2} \alpha_b b = 0$. We may split this up as

$$\sum_{b \in B_1 \setminus B} \alpha_b b = - \sum_{b \in B_2} \alpha_b b.$$

The left hand side lies in W_1 while the right hand side lies in W_2 , so both sides lie in $W_1 \cap W_2$. Since B is a basis for $W_1 \cap W_2$ contained in both B_i s, uniqueness shows that $\alpha_b = 0$ for all $b \in B_1 \setminus B$ and $b \in B_2 \setminus B$. Thus, $\sum_{b \in B} \alpha_b b = 0$, and so linear independence shows that $\alpha_b = 0$ for all $b \in B$ as well. Thus, $B_1 \cup B_2$ is linearly independent, as desired. ■

Working with finite-dimensional vector spaces allows us to perform arithmetic with dimensions.

Theorem 4.2.2 (Rank-Nullity). *Let $W \leq V$ be finite-dimensional vector spaces over k . Then*

$$\dim_k(V/W) = \dim_k V - \dim_k W.$$

In particular, V/W is finite-dimensional.

Proof. Let $S \subseteq W$ be a basis of W , and extend S to a basis T of V . We'll show

$$\{t + W : t \in T \setminus S\}$$

is a basis of V/W . For any $v \in V$, we may write $v = \sum_{t \in T} \alpha_t t$. Then

$$v + W = \sum_{t \in T} \alpha_t t + W = \sum_{s \in S} (\alpha_s s + W) + \sum_{t \in T \setminus S} (\alpha_t t + W) = \sum_{t \in T \setminus S} \alpha_t (t + W),$$

so this set is spanning. Now, if $\sum_{t \in T \setminus S} \beta_t (t + W) = 0 + W$, then $\sum_{t \in T \setminus S} \beta_t t \in W$, say $\sum_{t \in T \setminus S} \beta_t t = \sum_{s \in S} \beta_s s$, since S is a basis of W . Since T is linearly independent, we have that $\beta_t = 0$ for all $t \in T$, so this set is linearly independent. Thus, it forms a basis, as desired. ■

Corollary 4.2.3. *Let R be an integral domain and $N \leq M$ be R -modules. Then $\text{rk}_R(M/N) = \text{rk}_R M - \text{rk}_R N$.*

Proof. We've seen that

$$\text{rk}_R M = \dim_{\text{Frac } R}(R \setminus \{0\})^{-1} M.$$

Since localization commutes with quotients $()$, Rank-Nullity gives the result. ■

Corollary 4.2.4. *If $W \leq V$ are finite-dimensional vector spaces over k with $\dim V = \dim W$, then $V = W$.*

Proof. We have that $\dim V/W = \dim V - \dim W = 0$, so that $V/W = 0$ and $V = W$. ■

Rank-Nullity is often stated by using the first isomorphism theorem: if $T : V \rightarrow W$ is a linear transformation, then $V/\ker T \cong \text{im } T$, and so $\dim \text{im } T = \dim V - \dim \ker T$. Thus,

$$\dim \text{im } T + \dim \ker T = \dim V.$$

The value $\dim \text{im } T$ is often called the *rank* of T , while $\dim \ker T$ is called the *nullity*. This formulation gives us the following.

Corollary 4.2.5. *If V and W are finite-dimensional vector spaces over k and $T : V \rightarrow W$ is a linear transformation, then the following are equivalent.*

- (1) T is an isomorphism
- (2) T is injective
- (3) T is surjective

Proof.

(1) $\Rightarrow$ (2) Clear.

(2) $\Rightarrow$ (3) We have that

$$\dim W = \dim V = \dim \text{im } T + \dim \ker T = \dim \text{im } T,$$

so that $W = \text{im } T$.

(3) $\Rightarrow$ (1) We have that

$$\dim W = \dim V + \dim \text{im } T + \dim \ker T = \dim W + \dim \ker T,$$

so that $\dim \ker T = 0$. Thus, $\ker T = 0$, so T is injective and surjective, thus an isomorphism. ■

4.2.2 Nilpotent Operators

Recall that an element a of a ring R is *nilpotent* if $a^m = 0$ for some $m > 0$. We'll say a linear transformation $T : V \rightarrow V$ is nilpotent if it is nilpotent as an element of $\text{End}(V)$: that is, if $T^m = 0$ for some $m > 0$. Our main tool in tackling nilpotent operators is the following.

Proposition 4.2.6. *Let V be a finite-dimensional vector space over k , and suppose that $\dim V = n$. Then V is both Noetherian and Artinian. Moreover, every ascending (or descending) chain of subspaces contains at most $n + 1$ distinct subspaces.*

Proof. It suffices to prove the second statement. Suppose we have a chain

$$V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_n \subsetneq V_{n+1} \subseteq V.$$

Since there are proper inclusions at each step, we must have that $\dim V_i > \dim V_{i-1}$, so that $\dim V_i \geq \dim V_{i-1} + 1$. Thus, $\dim V_{n+1} \geq n + 1 + \dim V_0 > n$, contradicting the fact that V is n -dimensional. ■

Remark 4.2.7. The above could equivalently be shown by finding a composition series of length $n + 1$ for V and applying ■

This bound on the number of distinct members of a chain allows us to nicely analyze kernels. The picture the reader should have in their minds in the following proposition is the chain

$$0 = \ker T^0 \subseteq \ker T^1 \subseteq \ker T^2 \subseteq \cdots$$

Proposition 4.2.8. *Let V be a finite-dimensional vector space over k , $T : V \rightarrow V$ be a linear transformation, and $n = \dim V$.*

- (1) *If $\ker T^m = \ker T^{m+1}$ then $\ker T^m = \ker T^{m+k}$ for all $k \geq 0$.*
- (2) *$\ker T^n = \ker T^{n+1}$.*
- (3) *$\ker T^n = \{v \in V : T^m v = 0 \text{ for some } m > 0\}$.*
- (4) *T is nilpotent if and only if $T^n = 0$.*

Proof.

- (1) We'll show that $\ker T^{m+k} = \ker T^{m+k+1}$ for all $k \geq 0$. Clearly we have $\ker T^{m+k} \subseteq \ker T^{m+k+1}$. Conversely, suppose $v \in \ker T^{m+k+1}$, so that $0 = T^{m+k+1}v = T^{m+1}(T^k v)$. Then $T^k v \in \ker T^{m+1} = \ker T^m$, so that $T^m(T^k v) = 0$. Thus, $v \in \ker T^{m+k}$, as desired.
- (2) If this is false, then by (1), we must have $\ker T^m \neq \ker T^{m+1}$ for all $m \leq n$, so we get a chain

$$0 = \ker T^0 \subsetneq \ker T^1 \subsetneq \cdots \subsetneq \ker T^n \subsetneq \ker T^{n+1}$$

of length $n + 2$, contradicting Proposition 4.2.6.

- (3) We have that $\{v \in V : T^m v = 0 \text{ for some } m > 0\} = \bigcup_{m>0} \ker T^m$. Since $\ker T^m = \ker T^n$ for $m > n$, this is equal to $\bigcup_{m=1}^n \ker T^m$. Since $\ker T^m \subseteq \ker T^n$ for $m < n$, this is equal to $\ker T^n$.
- (4) Clearly if $T^n = 0$ then T is nilpotent. Conversely, if T is nilpotent, then $\ker T^n = \{v \in V : T^m v = 0 \text{ for some } m > 0\} = V$.

■

The main content of this proposition is really that the chain

$$0 = \ker T^0 \subseteq \ker T^1 \subseteq \cdots \subseteq \ker T^n \subseteq \cdots$$

must stabilize by $\ker T^n$. This fact allows us to decompose V into T -invariant subspaces where we can easily describe the action of T .

Proposition 4.2.9. *Let V be an n -dimensional vector space over k and $T : V \rightarrow V$ be a linear transformation. Then*

$$V = \ker T^n \oplus \operatorname{im} T^n$$

and these are T -invariant subspaces. Further, the action of T is nilpotent on $\ker T^n$ and an isomorphism on $\operatorname{im} T^n$.

Proof. We first show that $\ker T^n \cap \operatorname{im} T^n = 0$. Indeed, suppose that $v = T^n w$ lies in this intersection. Then $0 = T^n v = T^{2n} w$, so that $w \in \ker T^{2n}$. By Proposition 4.2.8, $w \in \ker T^n$, and so $v = T^n w = 0$.

Now, Proposition 4.2.1 shows that

$$\dim(\ker T^n \oplus \operatorname{im} T^n) = \dim \ker T^n + \dim \operatorname{im} T^n,$$

which is equal to $\dim V$ by Rank-Nullity. Thus, $V = \ker T^n \oplus \operatorname{im} T^n$.

We now show these spaces are T -invariant. If $v \in \ker T^n$, then

$$T^n(Tv) = T(T^n v) = T0 = 0,$$

so $Tv \in \ker T^n$. Similarly, if $v = T^n w \in \operatorname{im} T^n$, then

$$Tv = T(T^n w) = T^n(Tw)$$

lies in $\operatorname{im} T^n$.

Now, for $v \in \ker T^n$, we have that $T^n v = 0$, so that $(T|_{\ker T^n})^n = 0$. That is, the action of T on $\ker T^n$ is nilpotent. To show that the action of T on $\operatorname{im} T^n$ is an isomorphism, it suffices to show that the action is injective by Corollary 4.2.5. We indeed have that

$$\ker T|_{\operatorname{im} T^n} = \ker T \cap \operatorname{im} T^n \subseteq \ker T^n \cap \operatorname{im} T^n = 0.$$

■

We note that the only fact we used in showing that these subspaces are T -invariant is that T commutes with T^n . Thus, with very little work, we can strengthen our result.

Corollary 4.2.10. *With notation as above, let $S : V \rightarrow V$ be a linear transformation such that $ST = TV$. Then $\ker T^n$ and $\operatorname{im} T^n$ are S -invariant. In particular, this is true if $T = f(S)$ for some polynomial f .*

4.2.3 Eigenvalues and Eigenvectors

Given a linear transformation $T : V \rightarrow V$, the eigenvalues and eigenvectors provide a canonical way to describe how T acts. For any $\lambda \in k$, we write

$$E_T(\lambda) = \{v \in V : Tv = \lambda v\}.$$

This is a subspace of V , as it is equal to $\ker(T - \lambda \operatorname{id}_V)$, and we call this the *eigenspace* of λ . It is also not hard to see that $E_T(\lambda)$ is T -invariant.

Definition 4.2.11. Let V be a vector space over k and $T : V \rightarrow V$ be a linear transformation. An element $\lambda \in k$ is said to be an *eigenvalue* of T if $E_T(\lambda) \neq 0$. Nonzero elements of $E_T(\lambda)$ are said to be *eigenvectors* of T with eigenvalue λ .

A best-case scenario would be getting a basis of V consisting of eigenvectors of T , since we would then easily be able to read off how T acts on any given vector: the corresponding matrix would be diagonal.

Definition 4.2.12. A linear transformation $T : V \rightarrow V$ is said to be *diagonalizable* if there exists a basis of V consisting of eigenvectors of T .

As the reader may guess, a general linear transformation is not diagonalizable. There are two ways diagonalizability could fail: the eigenvectors could be linearly dependent, or there could not be enough to span V . The following proposition shows that the latter is the only problem.

Proposition 4.2.13. *Let $T : V \rightarrow V$ be a linear transformation, $\lambda_1, \dots, \lambda_m$ be distinct eigenvalues of T , and $v_i \in E_T(\lambda_i)$. Then $v_1, \dots, v_m$ are linearly independent.*

Proof. We induct on m , with $n = 1$ being clear: v_1 is nonzero, and so if $\alpha_1 v_1 = 0$ then $\alpha_1 = 0$. Now, if we have a linear dependence relation $\alpha_1 v_1 + \dots + \alpha_m v_m = 0$, then applying T gives

$$\lambda_1 \alpha_1 v_1 + \dots + \lambda_m \alpha_m v_m = 0.$$

Since we also have that

$$\lambda_n \alpha_1 v_1 + \dots + \lambda_m \alpha_m v_m = 0,$$

subtracting gives

$$(\lambda_1 - \lambda_m) \alpha_1 v_1 + \dots + (\lambda_{m-1} - \lambda_m) \alpha_{m-1} v_{m-1} = 0.$$

By induction $(\lambda_i - \lambda_m) \alpha_i = 0$ for $1 \leq i \leq m-1$. Since $\lambda_i \neq \lambda_m$, we have that $\alpha_i = 0$ for all $i \leq m-1$, and so $\alpha_m = 0$ as well, as desired. ■

We see in particular that, T has at most $\dim V$ eigenvalues, and if T has exactly $\dim V$ eigenvalues, then it is diagonalizable (though note that this is not necessary, as an eigenvalue can have linearly independent eigenvectors). This also means that we have a direct sum

$$E_T(\lambda_1) \oplus \dots \oplus E_T(\lambda_m).$$

Thus, T will be diagonalizable if and only if this direct sum is equal to V .

So far, we've been considering vectors v with $(T - \lambda \operatorname{id}_V)v = 0$. We may want to consider vectors such that we have $(T - \lambda \operatorname{id}_V)^m v = 0$ for some $m > 0$. By Proposition 4.2.8, this is equal to $\ker(T - \lambda \operatorname{id}_V)^n$, where $n = \dim V$.

Definition 4.2.14. Let V be an n -dimensional vector space over k , $\lambda \in k$, and $T : V \rightarrow V$ be a linear transformation. The *generalized eigenspace* of λ is

$$G_T(\lambda) = \ker(T - \lambda \text{id}_V)^n.$$

Nonzero elements of $G_T(\lambda)$ are called *generalized eigenvectors* of T with eigenvalue λ .

Note that we always have that $E_T(\lambda) \subseteq G_T(\lambda)$ and that $E_T(\lambda) \neq 0$ if and only if $G_T(\lambda) \neq 0$. Just as with eigenvectors, generalized eigenvectors are linearly independent.

Proposition 4.2.15. Let V be an n -dimensional vector space over k , $T : V \rightarrow V$ be a linear transformation, $\lambda_1, \dots, \lambda_m$ be distinct eigenvalues of T , and $v_i \in G_T(\lambda_i)$. Then $v_1, \dots, v_m$ are linearly independent.

Proof. Suppose that $\sum_{i=1}^m \alpha_i v_i = 0$. Let $k \geq 0$ be the largest k such that $(T - \lambda_1 \text{id}_V)^k v_1 \neq 0$, and set $w = (T - \lambda_1 \text{id}_V)^k v_1$, so that w is nonzero and $Tw = \lambda_1 w$. Set

$$S = (T - \lambda_1 \text{id}_V)^k (T - \lambda_2 \text{id}_V)^n \cdots (T - \lambda_m \text{id}_V)^n,$$

so that $Sw_i = 0$ for $i \neq 1$. We have that

$$Sw_1 = (T - \lambda_2 \text{id}_V)^n \cdots (T - \lambda_m \text{id}_V)^n w = \prod_{i=2}^m (\lambda_1 - \lambda_i)^n w.$$

Thus,

$$0 = S \left(\sum_{i=1}^m \alpha_i v_i \right) = \alpha_1 \prod_{i=2}^m (\lambda_1 - \lambda_i)^n w.$$

Since the λ_i are distinct and $w \neq 0$, this means $\alpha_1 = 0$. There was nothing special about 1 in this proof, so $\alpha_i = 0$ for all i . ■

Thus, we also have a direct sum

$$G_T(\lambda_1) \oplus \cdots \oplus G_T(\lambda_m)$$

that contains $E_T(\lambda_1) \oplus \cdots \oplus E_T(\lambda_m)$. This means that, in order for T to be diagonalizable, we must have $E_T(\lambda) = G_T(\lambda)$ for all λ . This is not sufficient in general, but does suffice when k is algebraically closed, as we will soon show.

4.2.4 The Minimal Polynomial

If V and W are finite dimensional vector spaces over k , say $\dim V = n$ and $\dim W = m$, then we know $\text{Hom}_k(V, W) \cong \mathbb{M}_{m \times n}(k)$, so that $\dim \text{Hom}_k(V, W) = mn$. In particular, $\dim \text{Hom}_k(V, V) = n^2$. Thus, if $T : V \rightarrow V$ is a linear transformation, then $1, T, \dots, T^{n^2}$ are linearly dependent over k , which means there is a polynomial $f(x) \in k[x]$ such that $f(T) = 0$. Thus, the ideal

$$\{f(x) \in k[x] : f(T) = 0\} \subseteq k[x]$$

is nonzero. Since $k[x]$ is a PID, this allows us to define the following.

Definition 4.2.16. Let V be a finite-dimensional vector space over k , and $T : V \rightarrow V$ be a linear transformation. The *minimal polynomial* of T , denoted m_T , is the unique monic generator of $\{f(x) \in k[x] : f(T) = 0\}$.

By definition, if $f(T) = 0$, then $m_T \mid f$. This fact, while seemingly simple, is actually quite powerful.

Theorem 4.2.17. Let V be a finite-dimensional vector space over k , and let $T : V \rightarrow V$ be a linear transformation. Then λ is an eigenvalue of T if and only if $m_T(\lambda) = 0$. In particular, if k is algebraically closed, then T has an eigenvalue.

Proof. Suppose first that λ is an eigenvalue of T with eigenvector v . Then the action of T on $\langle v \rangle$ is the same as scaling by λ , so that $m_{T|_{\langle v \rangle}}(x) = x - \lambda$. Since $m_T(T) = 0$, we also have that $m_T(T|_{\langle v \rangle}) = 0$. Thus, $x - \lambda \mid m_T$, so that λ is a root of m_T .

Conversely, suppose that $m_T(\lambda) = 0$, and write $m_T(x) = (x - \lambda)f(x)$. By the definition of m_T , we must have that $f(T) \neq 0$, say $f(T)v \neq 0$. Then

$$0 = m_T(T)v = (T - \lambda)(f(T)v),$$

so that $f(T)v$ is an eigenvector of T with eigenvalue λ . ■

In light of this result, we may hope that T is diagonalizable if m_T splits. Unfortunately, this is not true, but we can get a result that is still quite nice.

Theorem 4.2.18. *Let V be a finite-dimensional vector space over k and $T : V \rightarrow V$ be a linear transformation. Then there is a basis B of V such that $[T]_B$ is upper triangular if and only if m_T splits in $k[t]$. Further, if $[T]_B$ is upper triangular, then λ appears as a diagonal entry of $[T]_B$ if and only if λ is a root of m_T .*

Proof. Suppose first that m_T splits. We induct on $n = \dim V$. Since m_T splits, it has a root λ . Thus, λ is an eigenvalue of T , say with corresponding eigenvector v . Since $m_T(T) = 0$, we have that $m_T(T/\langle v \rangle) = 0$, so that $m_{T/\langle v \rangle} \mid m_T$ splits. Thus, by induction, we can find a basis $\{b + \langle v \rangle : b \in B\}$ of $V/\langle v \rangle$ such that $[T/\langle v \rangle]_{B/\langle v \rangle}$ is upper triangular. By Proposition 4.1.12,

$$[T]_{\{v\} \cup B} = \begin{pmatrix} \lambda & * \\ 0 & [T/\langle v \rangle]_{B/\langle v \rangle} \end{pmatrix}$$

is upper triangular.

Conversely, suppose that we can find a basis B such that $[T]_B$ is upper triangular, and let $\lambda_1, \dots, \lambda_n$ appear along the diagonal. Then $([T]_B - \lambda_1) \cdots ([T]_B - \lambda_n)$ is upper triangular with 0s along the diagonal, and so

$$([T]_B - \lambda_1)^n \cdots ([T]_B - \lambda_n)^n = 0.$$

Thus, $m_T \mid (x - \lambda_1)^n \cdots (x - \lambda_n)^n$, and so in particular m_T splits.

The above argument also shows that if λ is a root of m_T then λ must appear along the diagonal. Conversely, suppose that λ_1 is not a root of m_T , and set $I = \{i \in [n] : \lambda_i \neq \lambda_1\}$. Then m_T divides $\prod_{i \in I} (x - \lambda_i)^n$ (since λ_1 is not a root). But the $(1, 1)$ entry of $\prod_{i \in I} ([T]_B - \lambda_i)^n$ is $\prod_{i \in I} (\lambda_1 - \lambda_i)^n \neq 0$, a contradiction. Thus, λ_1 is indeed a root of m_T .³ ■

Corollary 4.2.19. *Nilpotent operators are triangularizable with 0s along the diagonal.*

Corollary 4.2.20. *Every operator over an algebraically closed field is triangularizable.*

Remark 4.2.21. In fact more is true: the multiplicity with which λ appears on the diagonal is always the same, and is equal to $\dim G_T(\lambda)$. The reader familiar with the characteristic polynomial will see this immediately, while the reader who is not will have to wait until we develop the theory of determinants in .

We can also strengthen the argument of the proof to see that $m_T \mid \prod_{i=1}^n (x - \lambda_i)$. Indeed, if $B = \{b_1, \dots, b_n\}$, we'll show by induction on $m \leq n$ that

$$(T - \lambda_1) \cdots (T - \lambda_m) b_i = 0$$

for all $i \leq m$. The case $m = 1$ is clear, since b_1 is an eigenvector of T with eigenvalue λ_1 . If this holds for $m - 1$ and $i < m$ then

$$(T - \lambda_1) \cdots (T - \lambda_m) b_i = (T - \lambda_m)((T - \lambda_1) \cdots (T - \lambda_{m-1}) b_i) = (T - \lambda_m) 0 = 0.$$

If $i = m$, then $(T - \lambda_m) b_m \in \langle b_1, \dots, b_{m-1} \rangle$, since $[T]_B$ is upper triangular. Since $(T - \lambda_1) \cdots (T - \lambda_{m-1})$ annihilates each of $b_1, \dots, b_{m-1}$ by induction, we see that $(T - \lambda_1) \cdots (T - \lambda_m) b_m = 0$. Taking $m = n$ shows that $(T - \lambda_1) \cdots (T - \lambda_n) = 0$.

We now turn to decomposing V based on the factorization of m_T .

Lemma 4.2.22. *Let V be an n -dimensional vector space over k , $T : V \rightarrow V$ be linear, and write $m_T = fg$ where f and g are relatively prime. Then $m_{T|_{\ker f(T)^n}} = f$. In particular, $\ker f(T)^n = \ker f(T)$.*

Proof. We must have that $m := m_{T|_{\ker f(T)^n}} \mid f^n, m_T$ so that it divides their gcd f . Suppose that $m \neq f$, so that mg properly divides m_T and $m(T)g(T) \neq 0$, say $m(T)g(T)v \neq 0$. Then

$$0 = m_T(T)v = f(T)g(T)v = f(T)(g(T)v),$$

so that $g(T)v \in \ker f(T) \subseteq \ker f(T)^n$. But $m(T)(g(T)v) \neq 0$ by construction, a contradiction. Thus, $m = f$. ■

Lemma 4.2.23. *Let V be an n -dimensional vector space over k , $T : V \rightarrow V$ be linear, and write $m_T = fg$ where f and g are relatively prime. Then $\text{im } f(T)^n = \ker g(T)$.*

³The reader will realize there is nothing special about 1 by recalling (or verifying) that diagonal entries multiply when dealing with upper triangular matrices.

Proof. We have that

$$\operatorname{im} f(T)^n \subseteq \operatorname{im} f(T) \subseteq \ker g(T).$$

To show equality, it suffices to show that $\ker g(T) \cap \ker f(T)^n = 0$ by Proposition 4.2.9, which is equal to $\ker g(T) \cap \ker f(T)$ by Lemma 4.2.22. Since f and g are relatively prime, we may write $1 = hf + \ell g$. If $v \in \ker g(T) \cap \ker f(T)$, then

$$v = h(T)f(T)v + \ell(T)g(T)v = 0.$$

■

Lemma 4.2.24. *Let V be an n -dimensional vector space over k , $T : V \rightarrow V$ be linear, and write $m_T = fg$ where f and g are relatively prime. Then $V = \ker f(T) \oplus \ker g(T)$ and $m_{T|_{\ker f(T)}} = f$, $m_{T|_{\ker g(T)}} = g$.*

Proof. By Proposition 4.2.9, $V = \ker f(T)^n \oplus \operatorname{im} f(T)^n$. By Lemma 4.2.22, $\ker f(T)^n = \ker f(T)$, and by Lemma 4.2.23, $\operatorname{im} f(T)^n = \ker g(T)$. Thus, $V = \ker f(T) \oplus \ker g(T)$. Lemma 4.2.22 shows the desired facts about minimal polynomials. ■

Theorem 4.2.25 (Primary Decomposition). *Let V be a finite dimensional vector space over k , $T : V \rightarrow V$ be linear, and $q_T = p_1^{e_1} \cdots p_r^{e_r}$, where the p_i are distinct irreducible polynomials. Then there exists a unique decomposition $V = V_1 \oplus \cdots \oplus V_r$ such that each V_i is T -invariant and $m_{T|_{V_i}} = p_i^{e_i}$.*

Proof. We first show existence by induction on r . Let $V_1 = \ker(p_1(T)^{e_1})$, which is T -invariant and has $m_{T|_{V_1}} = p_1^{e_1}$ by Lemma 4.2.24. Set $g = p_2^{e_2} \cdots p_r^{e_r}$. Then Lemma 4.2.24 gives

$$V = V_1 \oplus \ker g(T)$$

and $m_{T|_{\ker g(T)}} = g$, and so induction applied to $T|_{\ker g(T)}$ shows existence.

Now, given any such decomposition, we will show uniqueness by showing $V_1 = \ker p_1(T)^{e_1}$ (since there is nothing special about 1). Clearly $V_1 \subseteq \ker p_1(T)^{e_1}$. Write $W = V_2 \oplus \cdots \oplus V_r$, so that $m_{T|_W} = g$. Lemma 4.2.24 shows that $\ker g(T) \cap \ker p_1(T)^{e_1} = 0$. If $v \in \ker p_1(T)^{e_1}$ with $v = v_1 + w$ for $v_1 \in V_1$ and $w \in W$, we have that

$$0 = p_1(T)^{e_1}v = p_1(T)^{e_1}v_1 + p_1(T)^{e_1}w = p_1(T)^{e_1}w,$$

so that $w = 0$. Thus, $V_1 = \ker p_1(T)^{e_1}$, as desired. ■

Corollary 4.2.26. *If m_T splits, then V is the direct sum of the generalized eigenspaces of T .*

Corollary 4.2.27. *If m_T splits, then T is diagonalizable if and only if $E_T(\lambda) = G_T(\lambda)$ for all $\lambda \in k$.*

Corollary 4.2.28. *T is diagonalizable if and only if m_T splits with distinct roots.*